

First year Aerospace Engineering

Mathematics - Transformations

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1) Frequency and Period

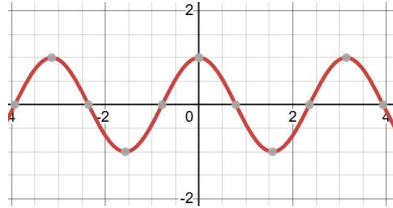
The frequency of the sine wave $y = \sin(x)$ is 1.
What is its period?

Will the period of the sine wave $y = \cos(2x)$ be larger or smaller? What about the frequency?

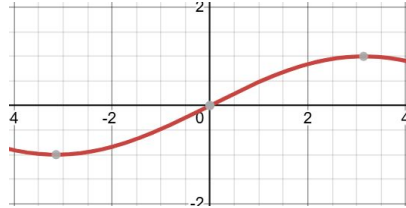
2) Frequency and Period

Which graph corresponds to which function?

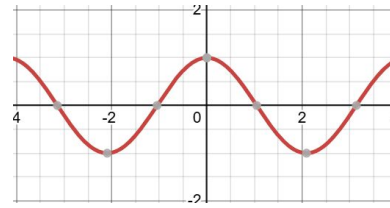
1)



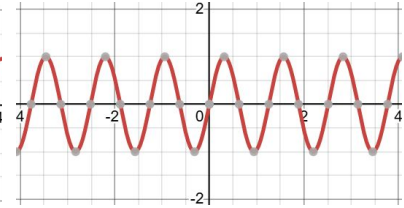
2)



3)



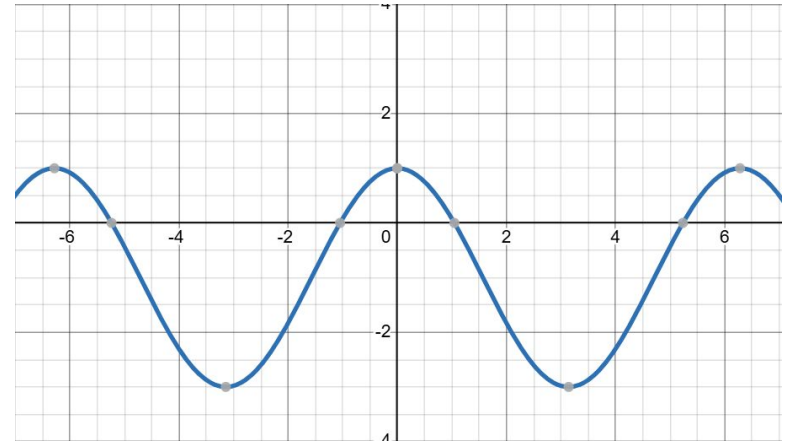
4)



- A) $y = \cos(3/2 x)$
- B) $y = \sin(5 x)$
- C) $y = \cos(2 x)$
- D) $y = \sin(1/2 x)$

3) Transformations

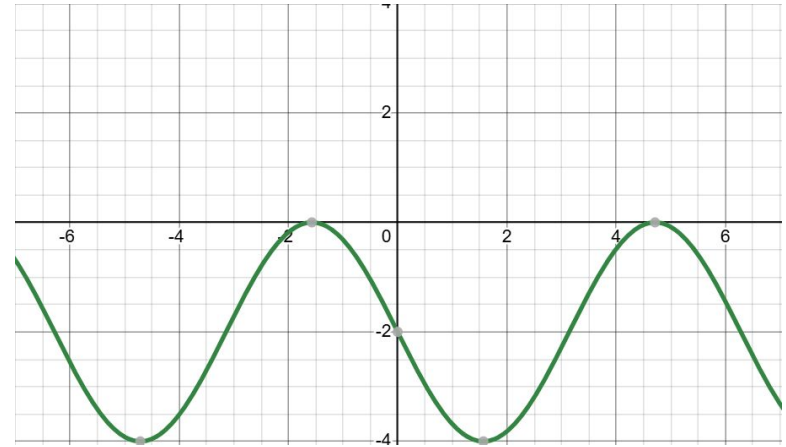
Which function corresponds to this graph?



- A) $f(x) = \cos(2x) - 2$
- B) $f(x) = -2(\sin(x) + 1)$
- C) $f(x) = 2\sin(x + \pi/2) - 1$
- D) $f(x) = -2\cos(x - \pi/2)$

4) Transformations

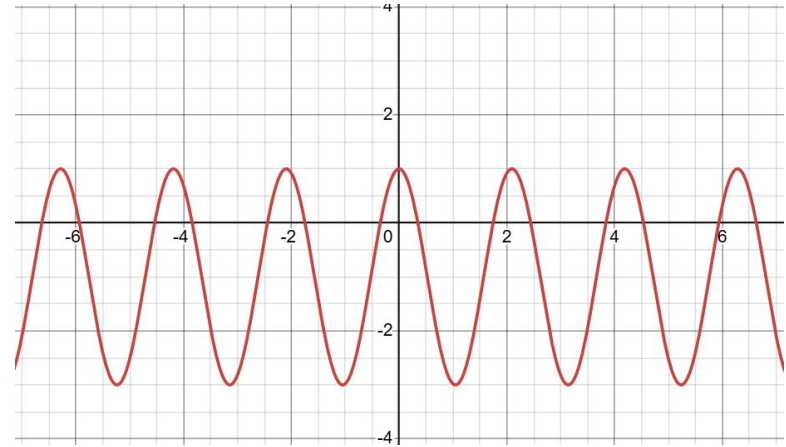
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5) Transformations

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6) Sketching

Plot the following functions in the range -2π to 2π

A) $y = 2 \cos\left(-2x - \frac{\pi}{2}\right) + 1$

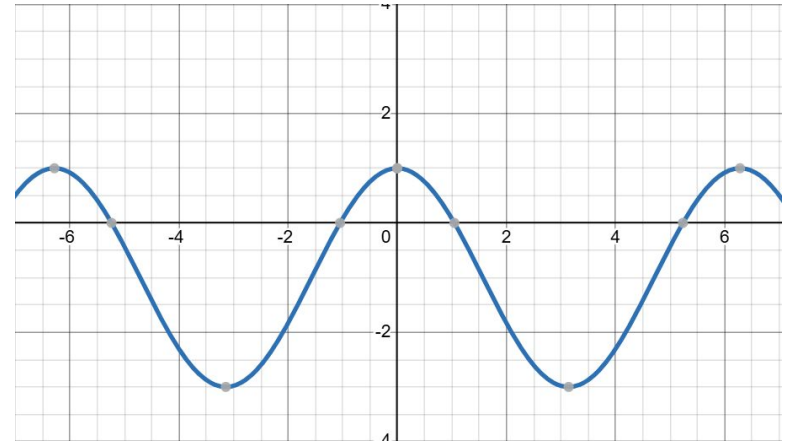
B) $y = 2 \cos(x) - \sin\left(x + \frac{\pi}{2}\right)$

C) $y = 6 \cos\left(-\frac{1}{3}x + \pi\right) - 2$

D) $y = \cos(x) + \sin\left(\frac{1}{2}x\right)$

3) Transformations

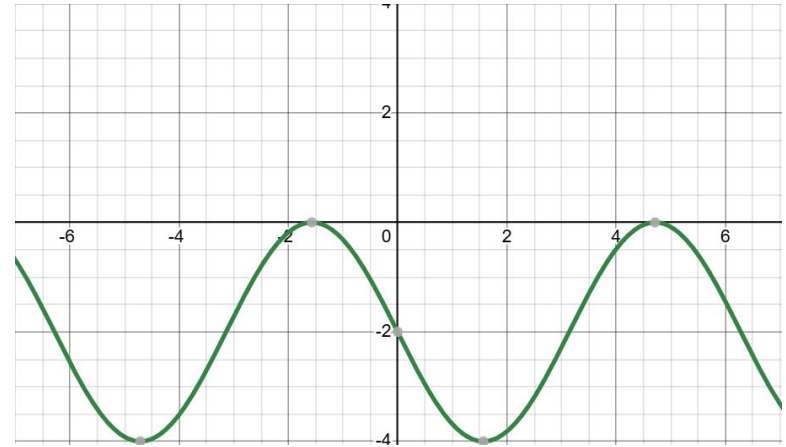
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4) Transformations

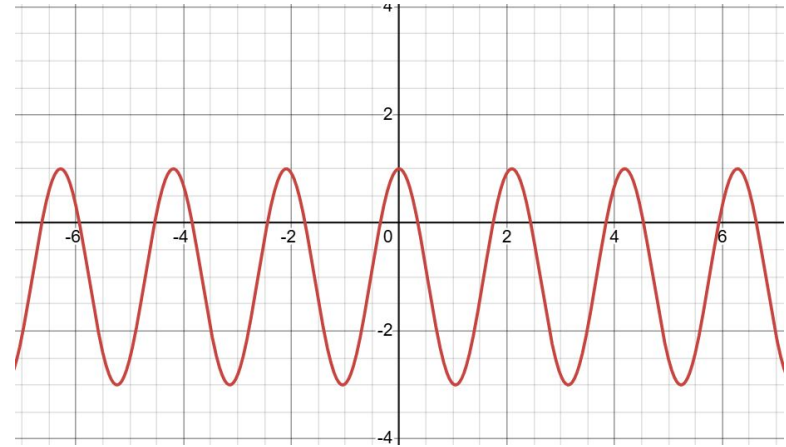
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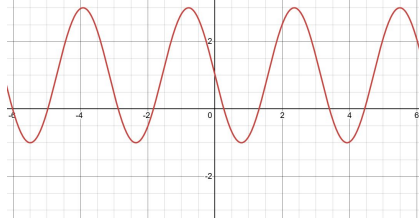


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- C) $f(x) = 2\sin(x + \pi/2) - 1$
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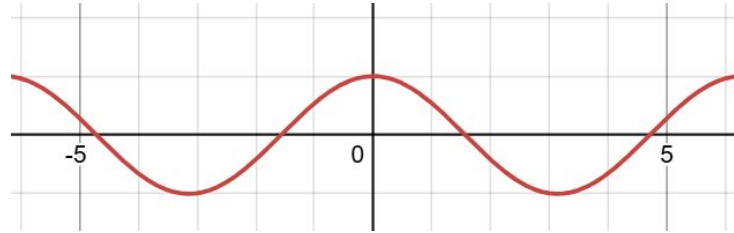
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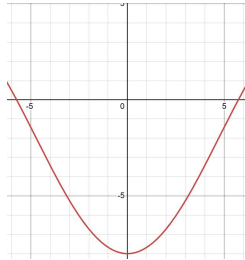
A)



B)



C)



D)

